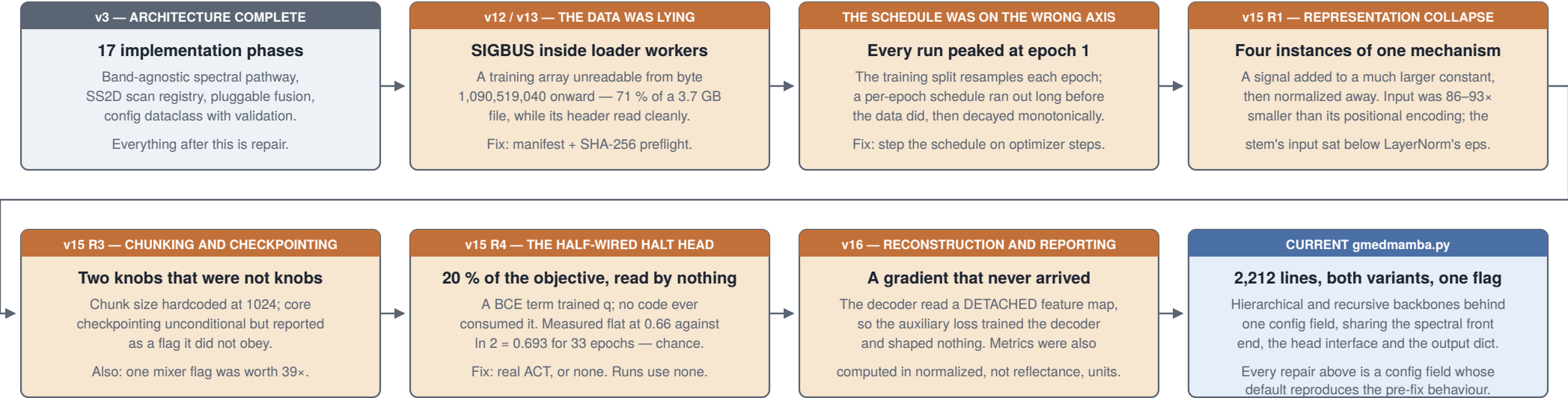


How the model file got here: seven revisions, and what each one found

Four of these defects produced no error. The code ran, the loss fell, the metrics were plausible, and something load-bearing was inert.



What caught them

Not review — measurement of the MECHANISM rather than the outcome:

- comparing the magnitude of a signal against the constant it is added to;
- checking whether a normalization's epsilon sits above or below the variance of its own input;
- asserting in a test that a gradient reaches a named module;
- reading what a term contributes to the objective against its own floor.

Several of these are now preflight gates that abort a run rather than annotate it. That is the form the lesson took in this repository.

The discipline that made the audit possible

Once a file is the audited entry point for a recorded run, it is not edited.

New capability arrives by subclassing or by wrapping at runtime:

- `prepare_histologyhsi_bc_v8` imports v6 and v7 — "frozen, imported not edited";
- `training/spectral_checkpoint.py` wraps at runtime so "gmedmamba.py stays untouched".

Where a change HAD to reach the model file, it became a config field defaulting to the old behaviour — so an existing checkpoint still loads with `strict=True`.

It costs duplication: eight `train_example_v*.py` and nine preparation scripts.

It buys the thing this paper needs: a run from five weeks ago is still rebuildable.